

• C. Olivocerebellar fibers synapse onto deep cerebellar nuclei directly

- D. Olivocerebellar fibers enter via the superior cerebellar peduncle

Answer: B

Explanation: Olivocerebellar fibers travel as climbing fibers and synapse directly onto Purkinje cells. All other afferents (spinocerebellar, pontocerebellar, vestibulocerebellar, etc.) synapse onto granule cells via mossy fibers. The inferior olive sends its axons via the inferior cerebellar peduncle, not the superior peduncle.

4. A 60-year-old man with a history of hypertension suddenly develops vertigo, nausea, vomiting, and difficulty walking. He also has ipsilateral facial numbness and contralateral body pain/temperature loss. Which artery is most likely occluded?

- A. Anterior inferior cerebellar artery (AICA)
- B. Superior cerebellar artery (SCA)
- C. Posterior inferior cerebellar artery (PICA)
- D. Basilar artery

Answer: C

Explanation: PICA is a branch of the vertebral artery and supplies the posterior-inferior cerebellum and lateral medulla. Occlusion causes lateral medullary (Wallenberg) syndrome: ipsilateral facial numbness, contralateral body pain/temperature loss, vertigo, nausea, vomiting, dysphagia, and Horner syndrome. AICA and SCA are branches of the basilar artery and supply different territories. Basilar occlusion would cause more widespread brainstem deficits.

5. During a neuroanatomy exam, a student is asked to identify the output pathway of the cerebellar hemispheres. The deep cerebellar nucleus that receives input from the cerebellar hemispheres (cerebrocerebellum) and projects to the thalamus via the superior cerebellar peduncle is the:

- A. Fastigial nucleus
- B. Interposed nucleus (globose and emboliform)
- C. Dentate nucleus
- D. Vestibular nucleus

Answer: C

Explanation: The dentate nucleus is the largest deep cerebellar nucleus and receives input from the lateral cerebellar hemispheres (cerebrocerebellum). It projects via the superior cerebellar peduncle to the contralateral thalamus and then to the motor cortex. The fastigial nucleus receives input from the vermis. The interposed nuclei receive input from the intermediate zone. The vestibular nucleus is associated with the vestibulocerebellum and is not a deep cerebellar

The vestibular nucleus is associated with the vestibulocerebellum and is not a deep cerebellar nucleus per se.

6. A 34-year-old woman presents with left-hand dysmetria and intention tremor that developed over 3-4 days. Her strength is 5/5 bilaterally and heel-to-shin is intact bilaterally. MRI shows a contrast-enhancing lesion in the left cerebellar peduncle. What is the most likely diagnosis?

- A. Ischemic stroke
- B. Multiple sclerosis
- C. Medulloblastoma
- D. Chronic alcoholism

Answer: B

Explanation: The patient's young age, female sex, and subacute progression over days are characteristic of MS. A stroke would present with sudden onset and minimal progression. Medulloblastoma is more common in children and typically presents with midline/vermis symptoms. Chronic alcoholism rarely causes focal cerebellar deficits, typically causing diffuse cerebellar atrophy affecting the vermis predominantly.

7. Which of the following best describes the somatotopic organization of the cerebellar cortex?

- A. The trunk is represented laterally and extremities medially
- B. The trunk is represented medially and extremities more laterally
- C. The head is represented in the vermis and limbs in the hemispheres
- D. There is no somatotopic organization in the cerebellum

Answer: B

Explanation: In the cerebellar cortex (particularly the spinocerebellum), the trunk/axial musculature is represented medially (in and near the vermis), while the extremities are represented more laterally. This mirrors the medial-to-lateral organization seen in the spinal cord and motor cortex. The head is represented in the anterior lobe of the vermis as well.

8. A patient presents with inability to perform rapid alternating movements of the hand, such as alternating pronation and supination. This sign is best described as:

- A. Dysmetria
- B. Intention tremor
- C. Dysdiadochokinesis

- D. Titubation

Answer: C

Explanation: Dysdiadochokinesis is the inability to perform rapidly alternating movements and is a sign of cerebellar hemispheric (lateral) dysfunction. Dysmetria refers to overshooting or undershooting a target. Intention tremor is a tremor that worsens as the limb approaches a target. Titubation is a rhythmic tremor of the head or trunk associated with midline/vermis lesions.

9. The cerebellar cortex receives two major types of afferent input. Which of the following correctly pairs the input fiber type with its primary neurotransmitter effect on the target neuron?

- A. Mossy fibers are inhibitory to granule cells
- B. Climbing fibers are excitatory to Purkinje cells
- C. Mossy fibers are inhibitory to Purkinje cells
- D. Climbing fibers are inhibitory to deep cerebellar nuclei

Answer: B

Explanation: Climbing fibers from the inferior olive are excitatory (glutamatergic) and synapse directly onto Purkinje cells, providing a powerful excitatory drive. Mossy fibers are also excitatory and synapse onto granule cells. Purkinje cells are the sole output neurons of the cerebellar cortex and are inhibitory (GABAergic) to the deep cerebellar nuclei.

10. A lesion of the right cerebellar hemisphere would most likely produce which of the following clinical findings?

- A. Left-sided dysmetria and intention tremor
- B. Right-sided dysmetria and intention tremor
- C. Bilateral truncal ataxia
- D. Left-sided weakness and hyperreflexia

Answer: B

Explanation: The cerebellum influences movement on the **IPSILATERAL** side. Therefore, a right cerebellar hemisphere lesion produces right-sided appendicular signs such as dysmetria, intention tremor, dysdiadochokinesis, and appendicular ataxia. Bilateral truncal ataxia is associated with midline/vermis lesions. Weakness and hyperreflexia suggest upper motor neuron (corticospinal) pathology, not cerebellar pathology.

11. Which of the following cerebellar peduncles carries the majority of cerebellar OUTPUT to

the thalamus and red nucleus?

- A. Inferior cerebellar peduncle
- B. Middle cerebellar peduncle
- C. Superior cerebellar peduncle
- D. Medial longitudinal fasciculus

Answer: C

Explanation: The superior cerebellar peduncle (brachium conjunctivum) is the primary efferent pathway from the cerebellum, carrying fibers from the deep cerebellar nuclei (especially dentate) to the contralateral thalamus (VL nucleus) and red nucleus. The inferior cerebellar peduncle carries spinocerebellar and olivocerebellar afferents. The middle cerebellar peduncle carries pontocerebellar afferents. The MLF is not a cerebellar peduncle.

12. A child presents with a midline posterior fossa tumor causing truncal ataxia and ataxic gait. Imaging shows a mass arising from the roof of the fourth ventricle. What is the most likely diagnosis?

- A. Glioblastoma multiforme
- B. Medulloblastoma
- C. Acoustic neuroma
- D. Meningioma

Answer: B

Explanation: Medulloblastoma is the most common malignant brain tumor in children and classically arises in the midline cerebellum (vermis/roof of 4th ventricle), causing midline cerebellar signs such as truncal ataxia and ataxic gait. Glioblastoma is most common in adults and typically occurs in the cerebral hemispheres. Acoustic neuromas arise from CN VIII. Meningiomas arise from the meninges.

13. The vestibulocerebellum receives input from the vestibular apparatus and projects output to the vestibular nuclei. This circuit is most important for controlling which of the following functions?

- A. Fine motor coordination of the fingers
- B. Planning of complex voluntary movements
- C. Balance, posture, and eye movements
- D. Somatosensory integration from the lower extremities

Answer: C

Explanation: The vestibulocerebellum (flocculonodular lobe) receives vestibular input and projects to the vestibular nuclei. It is critical for balance, posture, and coordination of eye movements. Fine motor coordination and movement planning are functions of the cerebrocerebellum (lateral hemispheres). Somatosensory integration from lower extremities is a spinocerebellum function.

14. A patient with cerebellar disease has intact muscle strength but is unable to coordinate the sequence of muscle contractions necessary to perform a smooth reaching movement, resulting in jerky, decomposed movements. This is best described as:

- A. Spasticity
- B. Asynergia
- C. Rigidity
- D. Chorea

Answer: B

Explanation: Asynergia (also called decomposition of movement) is the breakdown of smooth, coordinated multi-joint movements into a sequence of jerky, individual joint movements due to failure to properly coordinate synergistic muscle groups. It is a hallmark of cerebellar hemispheric dysfunction. Spasticity and rigidity are tone disorders. Chorea involves involuntary, random movements from basal ganglia dysfunction.

15. A neurologist notes that a patient's MRI shows atrophy predominantly of the anterior superior vermis. This pattern of cerebellar degeneration is most classically associated with which of the following conditions?

- A. Friedreich's ataxia
- B. Chronic alcoholism
- C. Multiple sclerosis
- D. Medulloblastoma

Answer: B

Explanation: Chronic alcoholism classically causes degeneration of the anterior superior vermis of the cerebellum, leading to gait ataxia and truncal instability with relative preservation of upper limb coordination. Friedreich's ataxia involves dorsal columns and spinocerebellar tracts. MS causes focal demyelinating lesions. Medulloblastoma is a tumor, not degenerative atrophy.

16. The spinocerebellar tracts transmit unconscious proprioceptive information to the cerebellum. The posterior spinocerebellar tract (PSCT) enters the cerebellum via which peduncle?

peduncle:

- A. Superior cerebellar peduncle
- B. Middle cerebellar peduncle
- C. Inferior cerebellar peduncle
- D. Medial longitudinal fasciculus

Answer: C

Explanation: The posterior spinocerebellar tract (PSCT) and cuneocerebellar tract enter the cerebellum via the inferior cerebellar peduncle. The anterior spinocerebellar tract (ASCT) enters via the superior cerebellar peduncle. The middle cerebellar peduncle carries pontocerebellar (corticopontocerebellar) fibers. The MLF is not a cerebellar peduncle.

17. A patient is unable to stand with feet together and eyes open or closed, swaying and nearly falling in both conditions. Which cerebellar region is most likely affected?

- A. Lateral cerebellar hemispheres bilaterally
- B. Flocculonodular lobe (vestibulocerebellum)
- C. Dentate nucleus
- D. Middle cerebellar peduncle

Answer: B

Explanation: The flocculonodular lobe (vestibulocerebellum) is essential for balance and equilibrium. Lesions cause severe balance problems that are present with both eyes open and closed (unlike sensory ataxia which worsens with eye closure - positive Romberg). Lateral hemisphere lesions cause appendicular ataxia. The dentate nucleus lesion would cause ipsilateral limb ataxia. Middle peduncle lesions affect input.

18. The corticopontocerebellar pathway relays information from the cerebral cortex to the cerebellum. What is the sequence of this pathway?

- A. Cerebral cortex → inferior olive → climbing fibers → Purkinje cells
- B. Cerebral cortex → pontine nuclei → middle cerebellar peduncle → granule cells
- C. Cerebral cortex → red nucleus → superior cerebellar peduncle → Purkinje cells
- D. Cerebral cortex → thalamus → inferior cerebellar peduncle → granule cells

Answer: B

Explanation: The corticopontocerebellar (pontocerebellar) pathway travels: cerebral cortex (frontal, occipital, temporal) → ipsilateral pontine nuclei → cross midline → middle cerebellar peduncle → cerebellar cortex granule cells (via mossy fibers). This pathway provides the cerebellum with information about motor intention. The inferior olive sends climbing fibers. The

superior peduncle is primarily efferent.

19. Which statement about Purkinje cells is correct?

- A. They are excitatory neurons that project to the cerebral cortex
- B. They are the sole output neurons of the cerebellar cortex and are GABAergic (inhibitory)
- C. They receive input exclusively from climbing fibers
- D. They project directly to motor neurons in the spinal cord

Answer: B

Explanation: Purkinje cells are the sole output neurons of the cerebellar cortex. They are GABAergic and therefore inhibitory. They project to and inhibit the deep cerebellar nuclei (and vestibular nuclei in the case of the vestibulocerebellum). They receive input from both climbing fibers (inferior olive) and parallel fibers (axons of granule cells, which are activated by mossy fibers). They do not project directly to the cortex or spinal cord.

20. A 55-year-old man presents with a sudden-onset severe headache, vomiting, vertigo, and inability to walk. His gait is severely ataxic and he falls to the right. MRI shows an infarct in the right posterior inferior cerebellum. Which artery is most likely occluded?

- A. Right anterior inferior cerebellar artery (AICA)
- B. Right superior cerebellar artery (SCA)
- C. Right posterior inferior cerebellar artery (PICA)
- D. Right middle cerebral artery

Answer: C

Explanation: PICA (posterior inferior cerebellar artery) is a branch of the vertebral artery that supplies the posterior-inferior cerebellum and lateral medulla. Occlusion causes severe headache, vertigo, nausea/vomiting, and ipsilateral cerebellar signs. AICA supplies the anterior-inferior cerebellum. SCA supplies the superior cerebellum. MCA supplies the lateral cerebral hemisphere and would not cause isolated cerebellar signs.

21. A right-handed 40-year-old woman undergoes testing of rapid alternating movements. She can perform rapid hand movements normally with her right hand but not with her left. She has 5/5 strength bilaterally. Where is the lesion?

- A. Right cerebral motor cortex
- B. Left cerebral motor cortex
- C. Right cerebellar hemisphere

Downloaded from <https://www.studocu.com>

24. During a neurological exam, a physician asks a patient to walk heel-to-toe in a straight line (tandem gait). The patient staggers and cannot maintain balance. This test is most useful for detecting lesions in which cerebellar region?

- A. Lateral cerebellar hemispheres
- B. Dentate nucleus
- C. Vermis/midline cerebellum
- D. Superior cerebellar peduncle

Answer: C

Explanation: Tandem gait testing (heel-to-toe walking) is highly sensitive for midline/vermis cerebellar lesions, which cause truncal ataxia and gait instability. Lateral hemispheric lesions primarily cause appendicular (limb) ataxia detected on finger-nose testing and dysdiadochokinesis. The dentate nucleus is the output nucleus of the lateral hemispheres. Superior cerebellar peduncle lesions would interrupt cerebellar output.

25. A neurology resident is asked to explain why a patient with a right cerebellar hemisphere lesion does not have weakness. Which statement best explains this observation?

- A. The cerebellum does not directly initiate or execute movement; it modulates and coordinates ongoing motor commands from the motor cortex
- B. The cerebellum has redundant pathways that compensate for unilateral lesions
- C. The cerebellum only controls involuntary movements, not voluntary motor strength
- D. Cerebellar lesions cause weakness only when both hemispheres are affected

Answer: A

Explanation: The cerebellum does not directly initiate movement or control muscle strength. It receives sensory and motor information, compares motor intention to motor output, and adjusts/modulates ongoing movements for coordination, timing, and accuracy. The actual motor commands for strength come from the corticospinal and other descending motor tracts. Therefore, cerebellar lesions produce incoordination (ataxia) without paralysis or weakness.
